BETWEEN REGULATION AND TRUST: THE ETHICAL DILEMMA IN DECENTRALIZED FINANCE

Alejandro Hernandez Ruthvik Penubarthi Smithi Saladi

ENGW3302, Professor Grasso
27 October, 2023

ABSTRACT

This literature review analyzes the ethical dilemmas of designing decentralized financial (DeFi) systems that operate outside of traditional regulatory frameworks created to protect consumers. Three major areas of concern are: (1) the conflict between institutional adaptation and decentralized autonomy, (2) the increased record of consumer harm caused by technical failures and fraud, and (3) unanswered questions of accountability and jurisdiction. empirical studies on DeFi-related losses show that current systems fail to meet the minimum standards of protection. However, existing frameworks such as New Institutional Economics, Dynamic Capabilities Theory, and Value-Sensitive Design indicate that ethical governance must integrate institutional reform with built-in design responsibility. This review finds that ethical DeFi development necessitates integrating transparency, accountability, and consumer safeguards directly into technical architecture in lieu of treating regulation as an external and separate constraint.

Introduction

Decentralized finance or DeFi uses blockchain technology to provide financial services without traditional banks or regulators. These systems operate through computer code or public blockchain, allowing anyone to participate without permission. Decentralization seeks to remove centralized authority, but it still relies on collective trust and governance parameters.

The current regulatory landscape for DeFi reveals a deep tension between technical autonomy and social responsibility, exposing a gap between ethical theory and how regulation functions in practice. This raises a critical ethical question: is it ethical to create financial systems outside regulatory frameworks designed to protect consumers?

The inquiry is vital because DeFi affects multiple stakeholders with competing interests. Consumers seek better financial opportunities but face unprecedented risks. Developers believe they are building fairer systems through code. Cypherpunks champion financial freedom from government control. Traditional banks view DeFi as both a threat and opportunity. Each group has different perspectives on whether regulation is necessary or antithetical to DeFi's purpose.

This paper examines the ethics of unregulated DeFi through three research avenues. First, we explore theoretical frameworks for regulating decentralized systems. Second, we document actual consumer harms from technical vulnerabilities, fraud and systemic failures. Third, we analyze accountability and jurisdiction challenges in systems without central authorities. Understanding these dimensions is crucial as governments worldwide develop regulations that will determine whether DeFi becomes a tool for financial inclusion or exploitation.

1. Theoretical Frameworks & Approaches to Regulation

A major ethical problem in decentralized finance (DeFi) is the challenge of regulating a system designed to avoid regulation. Governing DeFi ethically requires a combination of institutional theory and engineering-based design ethics to reconcile the freedom of decentralization while maintaining responsibility.

1A. Theoretical Frameworks

There are two dominant frameworks for understanding the ethical regulation of decentralized finance: New Institutional Economics (NIE) and Dynamic Capabilities Theory (DCT) (Liebau, 2025.) The NIE framework shows DeFi as a system that aims to minimize financial transaction costs and information gaps that exist in traditional finance by replacing intermediaries. This framework also highlights some potential risks of oversight and accountability, which require regulation that can adapt to decentralized environments. Hence, Liebau applies the concept of institutional isomorphism (DiMaggio & Powell, 1983) to argue that regulatory pressures will push DeFi toward hybrid models that integrate decentralized systems within established financial frameworks. DCT builds on this by emphasizing the institutional ability to sense opportunities, seize them through strategic action, and transform internal systems in response to new technologies and regulations (Teece, 2007).

Research on financial institutions emphasizes that social practices, like collaboration and collective learning, determine the success of digital adaptation and reinforces the idea that

these competencies must rely on shared norms of transparency, responsibility, and continuous ethical introspection (Schula & Som, 2024; Dong & Blankson, 2025; Jones et al., 2024). These findings, along with NIE and DCT, demonstrate that ethical DeFi governance requires institutions to develop new structures that incorporate transparency and accountability.

1B. Ethics and Design Integration

Resolving DeFi's ethical problems requires more than adapting institutions or regulations, it calls for embedding ethics directly into system design, a view supported by the field of blockchain technology ethics, which argues that ethical consistency should be integrated into system design rather than enforced externally (Ishmaev, 2025). This "responsibility by design" approach treats regulation as an engineering practice that combines Value-Sensitive Design (VSD) and Ethical Risk Analysis (ERA) to incorporate human values like fairness, privacy, and transparency into system design.

Further, ethical design should also reach beyond technical transparency to include financial literacy and informed participation, addressing the problem of asymmetrical information that often leaves users unaware of DeFi's risks (Liebau, 2025). Without this, ethical frameworks risk becoming a marketing tool, and self-regulation without fundamental changes could lead to "ethics washing," where private interests are prioritized above public benefits (Green, 2021). Together, these perspectives suggest that effective DeFi regulation should merge institutional adaptation with ethical design to create economic systems that are both functional and uphold human values.

1C. Regulatory Frameworks and Implementation

The European Union (EU) introduced the Markets in Crypto-Assets (MiCA) Regulation to increase transparency with standardized disclosure rules; however, this framework still faced difficulties in applying its rules to autonomous protocols with anonymous developers (Ferrari, 2020; Bajraktari, 2024). Moreover, effective regulation depends on the location where the harm occurs, which directly contradicts the digital nature of these transactions. This ongoing conflict between "code as law" and "law as governance" reinforces the idea that accountability should be built into decentralized systems (Ishmaev, 2025).

In response, EU regulators have begun exploring function-based regulation that focuses on financial activity rather than institutions (Bajraktari, 2024). This shift reflects the process of institutional isomorphism, (DiMaggio & Powell, 1983) in which regulators are being forced to adapt their frameworks to fit decentralized systems; collectively, this illustrates that regulation cannot rely solely on institutional adaptation or voluntary governance.

Although the EU currently classifies crypto-assets, investment tokens, and forms of electronic money under existing financial regulations (Ferrari, 2020), these measures have

proven insufficient for keeping pace with DeFi's evolving landscape. Likewise, because traditional enforcement tools such as licensing and asset freezing are ineffective against DeFi's code, the slow pace of legal adaptation continues to impede efforts aimed at maintaining security, service integrity, and payment oversight across member states (Bajraktari, 2024).

Additionally, analogical application of existing EU laws, specifically regarding data protection and smart contracts, revealed significant enforcement gaps and jurisdictional ambiguities (Graaf, 2019). Furthermore, the "privatization of money" through blockchain allows financial and political systems to be shaped by unaccountable actors, constituting an ethical urgency for developing regulatory frameworks for DeFi (Bajraktari, 2024.)

These issues underscore a need for more specific regulation that can address DeFi's transnational scope (Zetzsche, Arner, & Buckley, 2020), while balancing financial innovation with the consumer protection goals embedded in existing legislation (Ferrari, 2020), and closing recurring enforcement gaps across legal jurisdictions (Bajraktari, 2024). As a result, international coordination among regulators may become necessary, similar to the collaborative frameworks currently being discussed for cross-border payment systems. Importantly, this potential coordination strategy reflects aspects of Dynamic Capabilities Theory (Teece, 2007), where institutions sense and seize shared risks through internationally coordinated reform.

European agencies recognize this enforcement gap and are looking toward developing cross-border frameworks and auditing standards for decentralized systems (Bajraktari, 2024). Ultimately, the challenge lies in creating a legal framework that balances flexibility with accountability, ensuring that the benefits of decentralization do not come at the expense of stability and public trust.

1D. Ethical Implications

Hence, this regulatory uncertainty leaves everyday users exposed to financial harm. The ethical debate around regulation directly shapes real financial outcomes, so we can turn to consumer protection and documented harm to better understand these effects.

2. Consumer Protection & Documented Harms

DeFi systems present a fundamental tension between technological innovation and consumer protection. While proponents argue that code-based systems deliver fairer and more efficient financial services than traditional intermediaries, empirical evidence reveals systematic consumer harm at unprecedented scale. Between 2017 and 2022, DeFi systems generated approximately $9.3 billion in losses through 1,036 documented crime events, contributing to broader cryptocurrency industry losses of $30 billion (Carpentier-Desjardins et al., 2025). This massive wealth transfer from consumers to

malicious actors exposes critical vulnerabilities inherent in financial systems operating outside regulatory frameworks.

2A. Technical Vulnerabilities

The technical infrastructure of DeFi constitutes the primary mechanism through which consumer harm occurs. Technical vulnerabilities account for 52% of all attacks yet generate a disproportionate 83% of total financial damages (Carpentier-Desjardins et al., 2025), while smart contract exploits, particularly access control flaws, serve as the most common entry points for attackers (Qin et al., 2021).

Perhaps most troubling, the features marketed as DeFi's greatest innovations actually enable its most devastating attacks. Flash loans facilitate sophisticated exploits achieving returns exceeding 500% through pump-and-arbitrage schemes that can extract between $829,500 and $1.1 million within single atomic transactions (Qin et al., 2021). This paradox extends to governance mechanisms intended to democratize control, which instead create additional attack surfaces through vote buying and strategic token accumulation (Dotan et al., 2023). The negative correlation between voting rates and token prices identified suggests users remain unaware of these risks, even as vulnerabilities cascade both horizontally across interconnected protocols and vertically through governance layers.

2B. Evolution of DeFi Fraud

The evolution of fraud within DeFi shows a clear progression from opportunistic schemes to industrialized deception. The analysis of 101 documented abandonment schemes scams reveals that while early scams maintained operations for extended periods, later iterations operated for less than six months, indicating premeditated fraud from inception (Agarwal et al., 2024). This finding helps explain why DeFi actors perpetrate 41% of crime events but generate only 17% of financial damages (Carpentier-Desjardins et al., 2025).

The landscape consists of numerous small-scale schemes targeting retail investors alongside fewer but more devastating technical exploits achieving median losses of $2,073,280 per incident (Carpentier-Desjardins et al., 2025). These schemes employ consistent tactics including financial lures promising unrealistic returns, artificial time pressure, and authority mimicry impersonating legitimate projects (Agarwal et al., 2024). Fraud concentrates where information asymmetry peaks, with 73 of 101 abandonment schemes occurring through Initial Coin Offerings where technical complexity obscures malicious intent (Agarwal et al., 2024).

2C. Systemic Failure of DeFi

Systemic failures manifest through algorithmic mechanisms attempting to maintain stability without genuine economic backing. The Terra-Luna ecosystem's collapse from $18 billion to near-zero within eight days exemplifies how theoretical vulnerabilities

translate into catastrophic outcomes (Briola et al., 2022). The circular dependency between UST stablecoins and Luna tokens, combined with the Anchor protocol's unsustainable 20% yield absorbing 75% of UST's supply, created what functionally operated as a Ponzi scheme (Briola et al., 2022).

When initial de-pegging triggered selling pressure, stabilization attempts using Bitcoin reserves failed, initiating a death spiral that destroyed both tokens. This pattern mirrors the Iron Finance collapse, demonstrating that algorithmic stablecoins share identical failure modes: dependence on baseline demand that evaporates during crises, requirements for willing arbitrageurs who disappear when needed, and assumptions of informational efficiency that break down under stress (Briola et al., 2022).

2D. The Case for Regulatory Intervention

The documented harm establishes clear rationale for regulatory intervention. The market failures in DeFi systematically exceed those in traditional finance, with information asymmetries and systemic risks manifesting more severely in decentralized systems (Aquilina et al. 2024). This comparative analysis gains support from documentation of massive losses occurring without deposit insurance, supervision authorities, or systematic audits (Carpentier-Desjardins et al., 2025).

Proposed regulatory frameworks directly address identified vulnerabilities. The Markets in Crypto-Assets regulation requires legal entity status for issuers, detailed disclosures, and consumer rights including fund withdrawal (Agarwal et al., 2024). These requirements prevent anonymous teams from launching projects while targeting the deception patterns and information asymmetries that multiple studies identify as fundamental failures.

The convergence of evidence across technical analyses (Qin et al., 2021), crime statistics (Carpentier-Desjardins et al., 2025), fraud investigations (Agarwal et al., 2024), and systemic failure studies (Briola et al., 2022) reveals an ecosystem that systematically fails to protect consumers. The $9.3 billion in losses and 1,036 crime events represent not isolated incidents but systematic outcomes of choosing technological ideology over consumer welfare safeguards.

3. Accountability and Jurisdiction

DeFi systems face ethical challenges in two closely-related areas: accountability and jurisdiction. Accountability in decentralized finance refers to the assignment of legal and ethical responsibility when failures, misconduct, or financial harm occur, particularly determining which parties are liable for system failures (Schuler et al., 2024; Ferreira, 2024). Jurisdiction concerns the application and enforcement of regulatory frameworks across different countries and legal systems, which becomes particularly complex given DeFi's borderless, decentralized nature (Harvey & Rabetti, 2024; Ferreira, 2024).

Traditional financial systems operate within clear regulatory boundaries with defined responsible parties, but DeFi's distributed architecture challenges these established frameworks (Ferreira, 2024; Schuler et al., 2024).

3A. Accountability in DeFi Systems

The decentralized nature of DeFi makes accountability difficult to assign, as responsibility is dispersed among developers, platform operators, and users, with none holding clear legal or ethical liability (Schuler et al., 2024; Ferreira, 2024). This lack of defined accountability leaves regulators unsure of how to apply existing laws, since DeFi platforms rarely fit traditional financial models (Schuler et al., 2024; Charoenwong et al., 2025).

Developers often operate in gray areas without fully considering the ethics behind their design decisions (Schuler et al., 2024; Ferreira, 2024). DeFi platforms prioritize rapid deployment and growth over careful risk management, leaving safety and consumer protection as secondary concerns (Adamyk et al., 2025; Schuler et al., 2024). When problems arise– whether it be through hacking, scams, or protocol failures–there is no clear path to compensation. Users navigate systems where losses are irreversible and accountability remains too vague (Adamyk et al., 2025; Ferreira, 2024).

Decentralized governance mechanisms, such as token-based voting, cannot replace formal regulations or provide equivalent security to traditional financial systems (Schuler et al., 2024; Charoenwong et al., 2025). These self-governance tools are often limited in scope and are difficult to implement fairly (Schuler et al., 2024; Ferreira, 2024). The ideology behind DeFi shifts ethical responsibility from developers to users who may not be equipped to handle such risks (Ferreira, 2024; Harvey & Rabetti, 2024).

Meaningful accountability requires a shared framework where developers, operators, and regulators each take responsibility for ensuring transparency, compliance, and user safety (Schuler et al., 2024; Ferreira, 2024). The lack of proper safeguards and the many oversights causes users to face significant financial harm with minimal protection (Adamyk et al., 2025; Charoenwong et al., 2025).

3B. Jurisdictional Discrepancies within DeFi Systems

Regulatory frameworks vary significantly between countries, creating an inconsistent space where platforms may comply with standards in one jurisdiction while ignoring them in another (Harvey & Rabetti, 2024; Ferreira, 2024). Enforcement depends on where a platform operates and where harm occurs, but the lack of a centralized location and its borderless nature make it complicated to determine where blame should occur (Harvey & Rabetti, 2024; Schuler et al., 2024).

Cross-border transactions can bypass local rules, rendering current enforcement policies impractical (Ferreira, 2024; Harvey & Rabetti, 2024). Anti-money laundering (AML) compliance is particularly challenging without central authorities to monitor and regulate transactions (Adamyk et al., 2025; Schuler et al., 2024). Regulators struggle to keep pace with rapid technological developments of DeFi, leaving users potentially exposed to illicit schemes or financial crime (Charoenwong et al., 2025; Ferreira, 2024).

Even when platforms implement safeguards, these measures are often inconsistent and fail to cover all risks (Schuler et al., 2024; Adamyk et al., 2025). Without harmonized international regulations or enforceable rules, legal and compliance frameworks remain inadequate (Ferreira, 2024; Schuler et al., 2024). Users frequently remain unaware of vulnerabilities until experiencing losses (Harvey & Rabetti, 2024; Adamyk et al., 2025).

The combination of unclear accountability and fragmented jurisdiction creates a high-risk environment (Harvey & Rabetti, 2024; Ferreira, 2024). Until regulators, developers, and operators establish clear responsibilities and harmonized frameworks, DeFi users remain exposed to significant financial risks with minimal protection (Adamyk et al., 2025; Charoenwong et al., 2025).

Literary Gaps

Despite extensive documentation of DeFi's failures, critical gaps limit both understanding and solutions. Theoretical frameworks like Value-Sensitive Design are proposed as solutions, but no empirical studies test their implementation in functioning protocols. The literature identifies what should be done but provides no evidence of whether these approaches work in practice.

Studies document consumer losses but rarely examine how users understand risk or make decisions. Questions about cognitive biases, decision-making processes, and information requirements for informed participation remain unaddressed. Researchers identify that developers rarely face consequences, but no studies propose concrete accountability structures for decentralized development or explore how legal liability can attach to anonymous teams.

Research shows that pure decentralization fails to provide protection, yet few studies explore concrete hybrid models that balance autonomy with accountability. While individual protocol failures are well documented, research lacks comprehensive modeling of how cascading failures spread across interconnected ecosystems.

Finally, beyond documenting fraud, little research examines how protocol design incentivizes or discourages ethical behavior among developers and users. These gaps

reveal a literature stronger at diagnosis than prescription, suggesting future research must test specific governance models and protective architectures in live systems.

Discussion

Collectively, these gaps point to the need for frameworks that merge ethical design with adaptive regulation (Ishmaev, 2024; Liebau, 2025). The literature suggests that while DeFi innovation has outpaced oversight, its long-term viability still depends on the same legal stability and social trust that traditional finance provides (Liebau, 2025; Zetzsche et al., 2020). DeFi presents a foundational challenge to conventional regulatory strategies because it is designed to avoid intermediaries and centralized oversight while still requiring mechanisms of trust to function (Aquilina et al., 2024; Zetzsche et al., 2020).

Current regulatory models remain fundamentally mismatched to the decentralized nature of the systems they aim to govern (Bajraktari, 2024; Ferrari, 2020; Schuler et al., 2024). Across all three research avenues, the findings indicate that while existing frameworks can identify ethical and technical risks, they struggle to control them once authority becomes decentralized (Carpentier-Desjardins et al., 2025; Ferreira, 2024). As a result, DeFi's ethical challenges are not only technical but structural. Regulators and developers must move beyond viewing financial innovation and consumer protection as opposing goals and instead integrate them within shared governance systems (Liebau, 2025; Schuler et al., 2024).

The European Union's ongoing regulatory efforts illustrate this broader dilemma. Policies that promote financial innovation while acknowledging decentralized systems as potential threats to market stability and consumer protection still remain largely reactive to crises rather than preventative (Bajraktari, 2024; Ferrari, 2020). This difficulty in governing digital systems that operate beyond identifiable jurisdictions reflects a broader global tension between regulatory enforcement and proactive governance (Ferreira, 2024; Zetzsche et al., 2020).

The research also reveals a paradox central to decentralization. Despite removing intermediaries, DeFi systems still rely on centralized forms of trust in developers, auditors, or platforms (Aquilina et al., 2024; Schuler et al., 2024). In the absence of frameworks that integrate accountability and user literacy, ethical regulation remains functionally theoretical (Jones et al., 2024; Liebau, 2025). While the literature effectively identifies these risks, it offers few models capable of balancing user autonomy with long-term accountability (Carpentier-Desjardins et al., 2025; Ferreira, 2024). As a result, systems that operate without oversight create environments without recourse for harm and users remain vulnerable to exploitation, deepening public distrust in both decentralized and traditional institutions (Aquilina et al., 2024; Rettig et al., 2024).

On a structural level, the literature suggests a path forward centered on adaptation. First, regulation should shift from institution-based oversight that targets companies or intermediaries, to activity-based oversight that focuses on what financial functions a protocol performs regardless of who executes them (Aquilina et al., 2024; Bajraktari, 2024; Schuler et al., 2024). This approach reflects institutional isomorphism (DiMaggio & Powell, 1983), where regulatory pressures push DeFi toward hybrid models that integrate decentralized systems within established financial frameworks (Liebau, 2025; School, 2024).

Second, international coordination, through shared audit standards and integrated frameworks among regulators and developers, is essential in closing the jurisdictional gaps that make enforcement consistent (Bajraktari, 2024; Harvey & Rabetti, 2024; Zetzsche et al., 2020). This coordination strategy aligns with Dynamic Capabilities Theory (Teece, 2007), where institutions sense and seize shared risks through internationally coordinated reform (Dong & Blankson, 2025; Liebau, 2025; Schula & Som, 2024).

Third, mandating the integration of VSD and ERA can help integrate accountability directly into system architecture through audited smart contracts through practices of auditability, transparency and public education (Ishmaev, 2024; Liebau, 2025; School, 2024). This "responsibility by design" approach treats regulation as an engineering practice rather than an external constraint (Ishmaev, 2024), addressing the problem of asymmetrical information that often leaves users unaware of DeFi's risks (Jones et al., 2024; Liebau, 2025). Without fundamental changes, self-regulation could lead to "ethics washing," where private interests are prioritized above public benefits (Green, 2021).

Ultimately, the literature converges on a key insight that effective DeFi governance cannot rely solely on adapting existing legislation to new technologies (Ferrari, 2020; Ferreira, 2024; Schuler et al., 2024). The solution lies in an integration of ethics, engineering, and international cooperation to help create a coherent framework where trust is created by design (Ishmaev, 2024; Liebau, 2025).

Conclusion

The ethical challenges of decentralized finance stem from a conflict between technological innovation and consumer protection. This review shows that DeFi operates without adequate regulation, resulting in billions in losses while accountability remains unclear and consumer safeguards are absent.

Three areas of research point to the same finding: designing financial systems without protective mechanisms is an ethical failure, not a technological necessity. New Institutional Economics and Dynamic Capabilities Theory show that institutions must evolve to accommodate decentralization. However, the EU's Markets in Crypto-Assets

regulation demonstrates that traditional enforcement fails when applied to autonomous code. Value-Sensitive Design offers a path toward embedding human values into system architecture. Yet documented harms reveal the cost of inaction.

The jurisdiction gap exposes failed self-governance. Decentralized systems disperse responsibility among developers, operators, and users while giving unclear legal liability. Users face technical vulnerabilities they cannot understand, governance manipulation they cannot prevent, and financial losses they cannot recover. The ideology of decentralization has created systems that push risk onto those least able to manage it.

Ethical DeFi requires integrating transparency, accountability, and consumer protection directly into technical architecture. This means mandatory disclosure requirements, audited smart contracts, deposit insurance, and clear legal responsibility for protocol failures. Code is not neutral. Every design decision embeds values. The absence of safeguards is itself a choice that prioritizes innovation over safety. Without fundamental redesign, DeFi will continue extracting wealth from users while developers avoid consequences. Regulation must adapt to code-based systems, but developers must also build systems that protect users rather than exploit them.

References

Adamyk, B., Benson, V., Adamyk, O., & Liashenko, O. (2025). Risk management in defi: Analyses of the innovative tools and platforms for tracking defi transactions. Journal of Risk and Financial Management, 18(1), 38. https://doi.org/10.3390/jrfm18010038

Agarwal, S., Atondo-Siu, G., Ordekian, M., Hutchings, A., Mariconti, E., & Vasek, M. (2024). Short paper: Defi deception—uncovering the prevalence of rugpulls in cryptocurrency projects. In F. Baldimtsi & C. Cachin (Eds.), Financial Cryptography and Data Security (Vol. 13950, pp. 363–372). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-47754-6_21

Aquilina, M., Frost, J., & Schrimpf, A. (2024). Decentralized finance (Defi): A functional approach. Journal of Financial Regulation, 10(1), 1–27. https://doi.org/10.1093/jfr/fjad013

Bajraktari, H., & Qatani, V. (06 2024). Global Perspective: Legal Issues on DeFi Regulation in the European Union. https://doi.org/10.4018/979-8-3693-3346-4.ch007

Briola, A., Vidal-Tomás, D., Wang, Y., & Aste, T. (2023). Anatomy of a stablecoin's failure: The terra-luna case. Finance Research Letters, 51, 103358. https://doi.org/10.1016/j.frl.2022.103358

Carpentier-Desjardins, C., Paquet-Clouston, M., Kitzler, S., & Haslhofer, B. (2025). Mapping the DeFi crime landscape: An evidence-based picture. Journal of Cybersecurity, 11(1), tyae029. https://doi.org/10.1093/cybsec/tyae029

Charoenwong, B., Kirby, R. M., & Reiter, J. (2025). Tradeoffs in automated financial regulation of decentralized finance due to limits on mutable turing machines. Scientific Reports, 15(1), 3016. https://doi.org/10.1038/s41598-024-84612-9

DiMaggio, P. J., & Powell, W. W. (1983). The iron cage revisited: Institutional isomorphism and collective rationality in organizational fields. American Sociological Review, 48(2), 147. https://doi.org/10.2307/2095101

Dotan, M., Yaish, A., Yin, H.-C., Tsytkin, E., & Zohar, A. (2023). The vulnerable nature of decentralized governance in defi. Proceedings of the 2023 Workshop on

Decentralized Finance and Security, 25–31. https://doi.org/10.1145/3605768.3623539

Dong H, Blankson C (2025), Learning process challenges and outcomes in the fintech domain. *International Journal of Bank Marketing, Vol. 43 No. 8,* pp. 1601–1626 https://doi.org/10.1108/IJBM-05-2024-0263

Ferrari, V. (2020). The regulation of crypto-assets in the EU – investment and payment tokens under the radar. Maastricht Journal of European and Comparative Law, 27(3), 325-342. https://doi.org/10.1177/1023263X20911538

Ferreira, A. (2024). Decentralized finance (Defi): The ultimate regulatory frontier? Capital Markets Law Journal, 19(3), 242–259. https://doi.org/10.1093/cmlj/kmae007

Green, B. (2021). The contestation of tech ethics: A sociotechnical approach to technology ethics in practice. Journal of Social Computing, 2(3), 209–225. https://doi.org/10.23919/JSC.2021.0018

Harvey, C. R., & Rabetti, D. (2024). International business and decentralized finance. Journal of International Business Studies, 55(7), 840–863. https://doi.org/10.1057/s41267-024-00705-7

Ishmaev, G. (2024). Ethics of blockchain technologies. *SSRN (Social Science Research Network)*. http://dx.doi.org/10.2139/ssrn.5160243

Jones, M., Luu, T., and Samuel, B. (2024). Measuring Cryptocurrency Literacy. *SSRN (Social Science Research Network).* http://dx.doi.org/10.2139/ssrn.4541401

Liebau, D. (2025). Decentralized finance (Literacy) today and in 2034: Initial insights from Singapore and beyond. Emerging Markets Review, 69, 101378. https://doi.org/10.1016/j.ememar.2025.101378

Qin, K., Zhou, L., Livshits, B., & Gervais, A. (2021). Attacking the defi ecosystem with flash loans for fun and profit. In N. Borisov & C. Diaz (Eds.), Financial Cryptography and Data Security (Vol. 12674, pp. 3–32). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-662-64322-8_1

Rettig, R., Mosier, M., & Gilman, K. (2024). Genuine defi as critical infrastructure: A conceptual framework for combating illicit finance activity in decentralized finance. SSRN Electronic Journal. https://doi.org/10.2139/ssrn.4607332

School, G. L., Attorney, Capital Markets Law; J. S. M. ,. Stanford Law. (2024). Regulating Decentralized Financial Technology: A Qualitative Study on the Challenges of Regulating DeFi with a Focus on Embedded Supervision. Stanford Journal of Blockchain Law & Policy. https://stanford-jblp.pubpub.org/pub/regulating-defi

Schuler, K., Cloots, A. S., & Schär, F. (2024). On defi and on-chain cefi: How (Not) to regulate decentralized finance. Journal of Financial Regulation, 10(2), 213–242. https://doi.org/10.1093/jfr/fjad014

Teece, D. J. (2007). Explicating Dynamic Capabilities: The Nature and Microfoundations of (Sustainable) Enterprise Performance. *Strategic Management Journal*, *28*(13), 1319–1350. http://www.jstor.org/stable/20141992

Schula, U. and Som, O. (2024.) Digital Transformation in Financial Institutions: extending Dynamic Capabilities. *AMCIS 2024 Proceedings. 14.* https://aisel.aisnet.org/amcis2024/sig_osra/sig_osra/14

Zetzsche, D. A., Arner, D. W., & Buckley, R. P. (2020). Decentralized finance. Journal of Financial Regulation, 6(2), 172–203. https://doi.org/10.1093/jfr/fjaa010